

I can see you've shared a question about machine learning project lifecycle stages. Let me analyze each option for you.

Correct Answers: Data, Scoping, Modeling, and Deployment

Here's why each option applies or doesn't apply:

✓ Data

This is a fundamental stage. Every ML project requires data collection, cleaning, preprocessing, and exploration. Without proper data handling, models cannot be trained effectively.

✓ Scoping

This is typically the first stage where you define the problem statement, objectives, success metrics, and constraints. Understanding what you're trying to solve and why is essential before any technical work begins.

✗ Configuration

This is **NOT** typically recognized as a standard stage in mainstream ML lifecycle frameworks (like CRISP-DM, Google's ML framework, or industry standards). While configuration activities happen throughout a project (setting up environments, hyperparameters, etc.), it's not a named phase in the canonical lifecycle.

✓ Modeling

This is a core stage involving model selection, training, validation, and tuning. It's where you actually build and iterate on your machine learning algorithms.

✓ Deployment

This is the final major stage where you move the trained model into production, integrate it with applications, and make it available for real-world use. Post-deployment monitoring also typically falls here.

Common ML Lifecycle Framework Reference

Most frameworks follow something like this structure:

1. **Problem Definition/Scoping** → Understand what you're solving
2. **Data Collection & Preparation** → Get and clean the data
3. **Model Development/Modeling** → Build and train models
4. **Evaluation** → Test and validate performance
5. **Deployment** → Put into production
6. **Monitoring/Maintenance** → Track and update over time

The answer key for your question would be: **Data ✓, Scoping ✓, Modeling ✓, Deployment ✓** (with Configuration being the exception).